

Report
Of
STATIONERY SALES DATA ANALYTICS SUITE

Submitted
To
School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of
B.Tech CSE



By
Roll Number of Student: 2410030426
Name of Student: Shatakshi Bisht

Batch: 2024-2028

CANDIDATE'S DECLARATION

I, Shatakshi Bisht, do hereby declare that the internship report titled “Stationery Sales Data Analytics Suite” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering. All the references have been quoted and I have not taken as such any material from any other source. I affirm that this report is my own work and has not been submitted to any other institute for any degree/diploma requirement, and I shall be solely responsible for any kind of copyright violation in this regard.

Signature

Student Name: Shatakshi Bisht

Roll No.: 2410030426

ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluable constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.

I would like to thank Thiranex — Skill Development & Future Tech, for providing me the opportunity to work on this project under industry mentorship, and my faculty at the School of Computer Science and Engineering, IILM University, Greater Noida, for their continued guidance and support.

I express my gratitude to my parents for their blessings.

Signature

Student Name: Shatakshi Bisht

Roll No.: 2410030426

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INTERNSHIP COMPLETION CERTIFICATE

Certificate of Achievement issued by Thiranex — Skill Development & Future Tech



Internship Offer Letter



Date: 05 August 2026

TO,

Shatakshi Bisht

Intern ID: THX-AUG0526-492

Subject: Offer Letter for Internship in Data Analytics

Dear **Shatakshi Bisht**,

Following our selection process, we are pleased to offer you an internship position at **Thiranex**. We believe your skills and enthusiasm will be a great addition to our team.

Internship Role	Intern - Data Analytics
Commencement Date	05 Aug 2026
Completion Date	04 Sept 2026
Work Mode	Remote / Project-Based

During this tenure, you will work on practical projects under industry mentorship. Your progress will be reviewed periodically, and your performance will determine the successful completion of the internship.

We look forward to a mutually beneficial learning journey. Please confirm your acceptance by starting your onboarding tasks.



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For verification, contact Thiranex Verification Cell.
www.thiranex.in



PROJECT DESCRIPTION

4.1 Introduction

This report documents the work carried out during a one-month virtual internship in Data Analytics undertaken with Thiranex — Skill Development & Future Tech, from 05 August 2026 to 04 September 2026. The internship was remote and project-based, requiring the completion of four sequential tasks under industry mentorship, with progress reviewed periodically against fixed submission deadlines.

The internship project, titled “Stationery Sales Data Analytics Suite,” involved transforming a raw stationery sales dataset into actionable business insights through an interactive dashboard, customer segmentation, predictive forecasting, and automated data cleaning and reporting. The project provided hands-on exposure to the complete data analytics lifecycle — from data ingestion and cleaning to visualization, modelling, and automated reporting.

4.2 Organization Profile

Thiranex is a skill-development and future-tech organization that offers remote, project-based internships to students, enabling them to work on practical, industry-mentored projects across a defined tenure. Interns are assigned a structured sequence of tasks with fixed deadlines, and their performance on each task determines the successful completion of the internship.

- Organization: Thiranex — Skill Development & Future Tech
- Role: Intern — Data Analytics
- Work Mode: Remote / Project-Based
- Duration: 05 August 2026 – 04 September 2026

4.3 Problem Statement

How can raw stationery sales data be transformed into actionable business insights through interactive dashboards, customer segmentation, predictive forecasting, and automated data cleaning? The project addresses this question through four core deliverables: a Sales & Revenue Analysis Dashboard, a Customer Segmentation Project, Predictive Analytics using historical data, and Data Cleaning & Reporting Automation.

4.4 Project Objectives

- Design and implement an end-to-end solution for stationery sales data analysis.
- Build an interactive dashboard to visualize sales, revenue and KPI trends.
- Segment customers using clustering based on behaviour and demographics.
- Develop predictive models to forecast future sales trends.
- Apply data-cleaning techniques to handle missing values and duplicates.
- Automate reporting workflows to improve efficiency and consistency.

4.5 Scope of the Project

The scope of the Stationery Sales Data Analytics Suite is to develop an end-to-end data analytics solution that transforms raw stationery sales data into meaningful and actionable business insights. The project covers the complete process from data collection and cleaning to visualization, customer analysis, predictive forecasting, and automated reporting.

Task	Deliverable
1	Sales & Revenue Analysis Dashboard
2	Customer Segmentation Project
3	Predictive Analytics Using Historical Data
4	Data Cleaning & Reporting Automation

Table 4.1: Four-Task Project Journey

4.6 Technologies and Tools Used

Technology / Tool	Purpose
Python	Data analysis & scripting
Pandas / Excel	Data cleaning & processing
Power BI	Dashboards & KPI visualization
scikit-learn	Clustering for customer segmentation
Regression / Time-Series models	Predictive forecasting
GitHub	Version control & project delivery

Table 4.2: Technologies and Tools Used

4.7 System Architecture

The system follows a linear analytics pipeline. Raw Stationery sales data (CSV) is first passed through a data cleaning and automation stage, after which it feeds three parallel analytical components — the Sales & Revenue Dashboard, Customer Segmentation, and Predictive Analytics modules. The outputs of these components are consolidated into business insights and automated reports.

Stage	Description
1. stationery Sales Dataset (Raw CSV)	Raw transactional stationery sales data collected as the input source.
2. Data Cleaning & Automation	Missing values, duplicates and inconsistent records are handled and standardized.
3a. Sales & Revenue Dashboard	KPIs, revenue trends and top-performing products visualized with charts, filters and slicers.
3b. Customer Segmentation	Clustering applied to group customers by purchase behaviour and demographics.
3c. Predictive Analytics	Regression / time-series models forecast future sales trends.
4. Business Insights & Reports	Consolidated, automated reports and visual summaries for decision-making.

Figure 4.1: System Architecture of the stationery Sales Data Analytics Suite

4.8 Methodology

The internship followed a structured, week-wise methodology combining conceptual learning, hands-on labs, weekly assessments, and project submissions, as summarized in the table below

Step	Approach
1. Sales & Revenue Dashboard	Import data from Excel/CSV, visualize KPIs, and build interactive charts, filters and slicers.
2. Customer Segmentation	Perform clustering with Python (scikit-learn) / Power BI to group customers by behaviour and demographics.
3. Predictive Analytics	Apply regression / time-series models on cleaned historical data to forecast future sales trends.
4. Data Cleaning & Automation	Handle missing values and duplicates, then automate report generation and visual summaries.
5. Testing & Delivery	Validate each component independently, then track and submit deliverables against deadlines.

Table 4.3: Methodology — Task-wise Approach

Key implementation details of the dashboard and segmentation stages included:

- Importing sales data from Excel/CSV sources into a unified dataset.
- Visualizing KPIs — total sales, revenue trends, top-performing products — using charts, filters and slicers.
- Applying clustering (scikit-learn / Power BI) to segment customers and analyse purchase patterns and preferences.

Key implementation details of the predictive analytics and automation stages included:

- Using regression / time-series models for sales forecasting, and evaluating model accuracy.
- Cleaning and preprocessing historical datasets before modelling.
- Automating data cleaning with Python, Excel, or Power BI, and generating automated reports and visual summaries.

4.9 Expected Outcomes

The internship was expected to, and did, deliver the following learning outcomes:

- Data visualization, KPI tracking, and business insight generation.
- Hands-on experience in customer analytics, segmentation, and targeted insights.
- Practical exposure to predictive modelling, trend analysis, and forecasting.
- Understanding of data preprocessing, automation, and reporting efficiency.
- Improved skills in Python, Power BI/Excel, and end-to-end analytics workflows.
- Better connection between theoretical concepts and real-world data problems.

A few challenges were encountered and resolved during the project:

- Missing values and duplicates were resolved using automated Python/Pandas cleaning routines.
- Choosing the right clustering approach required testing multiple algorithms and features for segmentation.
- Balancing model accuracy against simplicity was addressed by comparing regression and time-series approaches.

Future scope identified for the project includes:

- Deploying the sales dashboard as a live, shareable web application.
- Exploring deep-learning models for more accurate demand forecasting.
- Integrating real-time / streaming data pipelines for continuous updates.
- Extending segmentation with RFM (Recency, Frequency, Monetary) analysis.

In summary, the Stationery Sales Data Analytics Suite delivered practical, end-to-end experience from data cleaning and dashboarding to customer segmentation and predictive forecasting along with structured, industry-mentored exposure to real project timelines and deliverables.

4.10 Certificates of Completion and Communication Proof

The following documents confirm the successful completion of the internship: the Internship Completion Certificate issued by Thiranex, and the official Offer Letter confirming the internship role, duration and terms.



Figure 4.2: Internship Completion Certificate

Figure 4.3: Offer Letter — Thiranex



SKILL DEVELOPMENT & FUTURE TECH

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TO,
Shatakshi Bisht
Intern ID: THX-AUG0526-492

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